Noah Niedzwiecki

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Skills

Programming: C, Bash, Python, SQL, PHP
Embedded Platforms: Embedded Linux, Yocto, U-Boot, systemd, Linux kernel bring-up
Networking & Data Models: TCP, UDP, YANG
Debugging & Performance: GDB, perf, flamegraph analysis, py-spy, strace
Tools: Git, Jira, tmux, Apache, AI-assisted development

Experience

Software Engineer, Ciena Corporation – Atlanta, GA — January 2020 – Present

- Led cross-functional delivery of an automatic fiber-optic transceiver software-upgrade feature by defining the high-level design, coordinating 10 developers across teams, and driving end-to-end release readiness.
- Served as technical prime for debug-symbol and performance-analysis tooling used by 100+ engineers, making core and trace debugging broadly accessible and reducing trace retrieval from undocumented, open-ended investigation to under 1 minute.
- Implemented a fiber-optic laser polarization state streaming pipeline with UDP relay and consumer components that transformed incoming data into YANG-defined structures for off-box consumption, with multi-stream and variable-rate support.
- Owned the software-management microservice lifecycle across multiple hardware targets, defining deployment and rollback behavior to standardize production feature and bug-fix releases.
- Led Jira triage and live customer incident response for the software-management microservice, diagnosing root causes and delivering fixes to restore service stability.
- Created a reusable Yocto recipe that bundled RPM sets into one deployable package and extended the software-management microservice with SHA-384 manifest validation, reducing load-validation time by 50%.
- Engineered a Bash- and tmux-based automation scripting framework with reusable shell-function libraries to automate SSH sessions, remote command execution, output monitoring, and complex test workflows.
- Upgraded remote-journal services by modernizing systemd integration and redesigning socket/service schemas, improving log-storage efficiency by 30% while validating max-use sizing under real-world workloads.
- Supported a security-driven major Linux kernel upgrade across multiple hardware platforms by compiling and validating builds, then debugging and fixing driver incompatibilities to enable stable security migration.
- Developed a web interface in HTML, JavaScript, and PHP that automated core-file stacktrace extraction and system-state analysis, including log triage, performance plotting, and boot-timing visualization.
- Created internal AI usage guidelines and taught practical workflows through team presentations, accelerating onboarding to modern development tools and increasing independent problem-solving across engineers.

Projects

Coffee Club — coffeeclub.us

- Built and deployed a production web application on a rented Linux server, installing dependencies, configuring systemd services, registering the domain, automating backups, and implementing secure JWT authentication, AES-256 data encryption, and SQL-backed persistence for real users.
- Developed full-stack platform features including REST API integration, SQL schema and query design, Google Maps support, and rate limiting while validating security and correctness in production.

Education

Auburn University – BS in Computer Engineering — December 2019